

1. Executive Summary

This report presents a concrete mix design performed in accordance with IS 10262:2019 (Indian). The design targets a characteristic compressive strength of 31.6 MPa with a water-cement ratio of 0.530.

2. Input Parameters

Design Standard

Code **IS 10262:2019 (Indian)**

Mix Parameters

Target Strength (f'c / fck) **25.0 MPa**

Required Slump **75.0 mm**

NMSA **20 mm**

Cement

Cement Type **OPC_43**

Specific Gravity **3.15**

Fine Aggregate

Specific Gravity **2.65**

Grading Zone **Zone II**

Coarse Aggregate

Specific Gravity **2.7**

IS-Specific Options

Exposure Class (IS 456) **None specified**

Target Volume **1.0 m^3**

3. Material Quantities

Proportions per Cubic Metre

- Cement **369.3 kg/m^3**
- Water **196.0 kg/m^3**
- Fine Aggregate (Sand) **645.6 kg/m^3**
- Coarse Aggregate **1169.5 kg/m^3**
- Total Aggregate **1815.1 kg/m^3**
- Water/Cement Ratio **0.530**
- Target Mean Strength (f'cr/ftm) **31.6 MPa**
- Air Content **1.0%**

4. Calculation Steps (Detailed)

Each step below shows the formula used, the input values, and the result. Clause references point to the relevant section of the design code.

Step 1: Target mean strength (ftm) [IS 10262:2009 Clause 4.1]

Formula: $f_{tm} = f_{ck} + 1.65 \times s$

Inputs: $f_{ck}=25.0$, $s=4.0$

Result: 31.60 MPa

Step 2: Water content [IS 10262:2009 Table 5]

Formula: From Table 5 by NMSA, slump, and grading zone

Inputs: $nmsa=20$, $slump=75.0$, $grading_zone=II$

Result: 196.00 kg/m³

Step 3: Water-cement ratio [IS 10262:2009 Figure 1/2/3]

Formula: From Figure for OPC_43 cement at f_{tm}

Inputs: $f_{tm}=31.6$, $cement_type=OPC_43$

Result: 0.53

Step 4: Cement content [IS 10262:2009 Clause 5.3]

Formula: $Cement = Water / W/C \text{ ratio}$

Inputs: $water=196.0$, $wc=0.5308$, $scm_replacement_pct=0$

Result: 369.25 kg/m³

Step 5: Coarse aggregate proportion [IS 10262:2009 Table 7]

Formula: $CA \text{ volume} = Total \text{ aggregate volume} \times Table \ 7 \text{ fraction}$

Inputs: $nmsa=20$, $grading_zone=II$, $ca_fraction=0.64$

Result: 1169.47 kg/m³

Step 6: Fine aggregate proportion [IS 10262:2009]

Formula: $FA \text{ volume} = Total \text{ aggregate volume} - CA \text{ volume}$

Inputs: $vol_total_agg=0.67677652181194$, $vol_ca=0.4331369739596416$

Result: 645.64 kg/m³

Step 7: Moisture-corrected water [IS 10262:2009]

Formula: $Adjusted \text{ water} = Design \text{ water} - \text{free moisture from aggregates}$

Inputs: $design_water=196.0$, $adjusted=208.30379716654105$

Result: 208.30 kg/m³

5. Warnings & Advisory Notes

! IS 10262:2009 recommends minimum 3 trial batches for validation

Note: The above warnings are advisory. The designer should review each item and confirm that the final mix is validated by trial batches before use in construction.

6. Engineering Context & Glossary

Target Mean Strength (ftm)

The target mean strength f_{tm} is calculated as:

$$f_{tm} = f_{ck} + 1.65 \times s$$

where f_{ck} is the characteristic compressive strength and s is the assumed standard deviation from IS 10262:2019 Table 1. This ensures that at least 95% of test results exceed f_{ck} .

Water/Cement Ratio

The W/C ratio determines concrete strength and durability. IS 10262 provides separate W/C ratio curves for OPC 33, OPC 43, and OPC 53 grade cements. The selected W/C is then checked against IS 456:2000 Table 3 exposure limits, which cap the W/C ratio based on environmental conditions.

Exposure Classes (IS 456:2000)

IS 456:2000 Table 5 defines exposure conditions that govern durability:

- * Mild: Min 220 kg/m³ cement, W/C ≤ 0.60
- * Moderate: Min 240 kg/m³ cement, W/C ≤ 0.60, grade M20
- * Severe: Min 250 kg/m³ cement, W/C ≤ 0.50, grade M25
- * Very Severe: Min 260 kg/m³ cement, W/C ≤ 0.45, grade M30
- * Extreme: Min 280 kg/m³ cement, W/C ≤ 0.40, grade M35

These limits ensure adequate durability for the intended service environment.

Grading Zones

Fine aggregate (sand) is classified into Grading Zones I-IV based on particle size distribution per IS 383. Zone II is the reference; coarser sand (Zone I) requires slightly less water, while finer sand (Zone III/IV) requires more. The water content is adjusted by -3% for Zone I and +3%/+6% for Zone III/IV.

Trial Batches

This mix design is a theoretical starting point. Before use in construction, the designer must prepare and test at least 3 trial batches to verify that the target strength and workability are achieved. Adjustments to water content, admixture dosage, or aggregate proportions may be needed based on trial results.

Moisture Correction

The aggregate quantities in this report are in saturated surface-dry (SSD) condition. In practice, the actual water added must be adjusted for the moisture content and absorption of the aggregates. If the aggregates are wetter than SSD, reduce the mixing water; if drier, increase it.

Volume Method

This design uses the absolute volume method. The volume of each ingredient (cement, water, air, aggregates) is calculated from its mass and specific gravity, and the total is verified to equal 1.0 m³. Any discrepancy is absorbed by adjusting the fine aggregate quantity.

Disclaimer

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